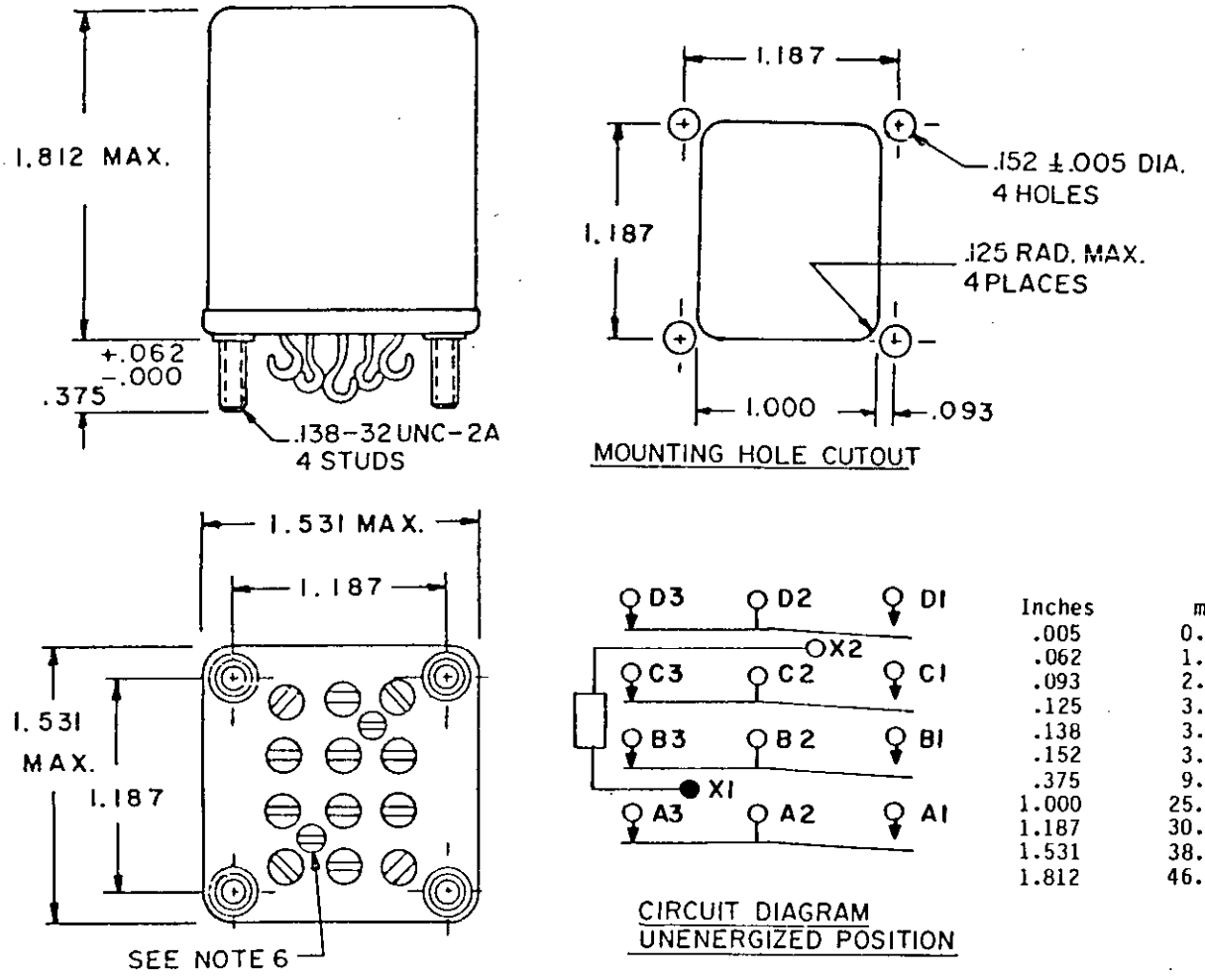
MILITARY SPECIFICATION SHEET

RELAYS, ELECTRICAL, HERMETICALLY SEALED, 4 PDT, 10 AMPERES

Ⓕ Inactive for new design after 12 April 1988.
For new design use MS25271.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-5757.



Inches	mm
.005	0.13
.062	1.57
.093	2.36
.125	3.18
.138	3.51
.152	3.86
.375	9.53
1.000	25.40
1.187	30.15
1.531	38.89
1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Shape of solder terminals optional.
5. There shall be affixed to the relay a legible circuit diagram which identifies each terminal location specified.
6. Terminal X1 shall be identified with either a contrasting glass bead or a colored dot adjacent to the terminal.

Ⓕ FIGURE 1. Dimensions and configuration.

Ⓕ denotes changes

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REQUIREMENTS:

Contact data:

Configuration: 4 PDT.

Arrangement: Four form C.

Load ratings (relay case grounded):

Resistive: 10 amperes at 28 V dc.
10 amperes at 115 V, 400 Hz, single phase.
10 amperes at 115/200 V, 400 Hz, three phase.
6 amperes at 115 V, 50/60 Hz.
6 amperes at 115/208 V, 50/60 Hz, three phase.

Inductive: 6 amperes at 28 V dc.
10 amperes at 115 V, 400 Hz, single phase.
10 amperes at 115/200 V, 400 Hz, three phase.
4 amperes at 115 V, 50/60 Hz.
4 amperes at 115/208 V, 50/60 Hz, three phase.

Lamp: 2 amperes at 28 V dc.
2 amperes at 115 V, 400 Hz, single phase.
2 amperes at 115/200 V, 400 Hz, three phase.
1.5 amperes at 115 V, 50/60 Hz.
1.5 amperes at 115/208 V, 50/60 Hz, three phase.

Minimum current: Applicable.

Contact resistance or voltage drop:

Rated life:

Before: 100 millivolts (mV) maximum.

During: 10 percent of open-circuit voltage maximum.

After: 200 mV maximum.

Minimum current:

Before: 0.05 ohm maximum.

During: 3 ohms maximum.

After: 1 ohm maximum.

Contact bounce: 2 milliseconds (ms) maximum.

Overload:

Resistive: Two times rated current.

Inductive: Two times rated current.

Coil data: (See table I):

Duty rating: Continuous.

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Ⓕ TABLE I. Part numbers and characteristics.

Part number M5757/18-1/	Maximum coil voltage	Rated coil voltage	Over the temperature range: Pickup voltage	Hold voltage	Dropout voltage	Coil resistance	Maximum operate time over temperature range	Maximum release time over temperature range	Magnetic interference	Shock
001	32 V dc	26.5 V dc	18 V dc	7 V dc	1.5 V dc	99 ohms minimum	25 ms	25 ms	Not applicable	Symbol 3 (100 g's)
003	122 V at 400 Hz	115 V at 400 Hz	90 V ac	30 V ac	5 V ac	2/ 2430 ohms (for reference only)	50 ms	50 ms	Applicable	Symbol 3 (100 g's)
005	122 V at 60 Hz	115 V at 60 Hz	90 V ac	30 V ac	5 V ac	2/ 3000 ohms (for reference only)	50 ms	50 ms	Applicable	Symbol 3 (100 g's)
007	122 V at 50/60 Hz	115 V at 50/60 Hz	90 V ac	30 V ac	5 V ac	2/ 3000 ohms (for reference only)	50 ms	50 ms	Applicable	Symbol 3 (100 g's)

1/ Part numbers M5757/18-002, -004, and -006 canceled without replacement.
2/ Maximum coil current: -003, 45 milliamperes, -005, 65 milliamperes, -007, 65 milliamperes.

Electrical data:

Insulation resistance: 1,000 megohms minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts:	1,000	250 all terminals to case
Between case, frame, or enclosure and coil:	1,000	
Between all contacts and coil:	1,000	
Between open contacts in the energized and unenergized positions:	1,000	
Between contact poles:	1,000	

Environmental data:

Temperature range: Symbol B (-65°C to +125°C).

Vibration: Symbol 3 (15 g's, 10 to 2,000 Hz).

Shock: See table I.

Coil life: Not applicable.

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Physical:

Terminal strength: 3 ±0.3 pound pull, coil terminals.

Sealed by welding: Not applicable.

Dimensions and configuration: See figure 1.

Termination: Solder lug.

Weight: 0.45 pound maximum.

(F) Life test requirements:

High level: 100,000 cycles at rated load.

Minimum current: Level II, 50,000 cycles.

Part number: M5757/18- (dash number from table I).

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
24 units plus 1 open unit qualification inspection as applicable	M5757/18-005	12 units plus 1 open unit. Qualification inspection as applicable.
	M5757/18-007	12 units plus 1 open unit. Qualification inspection as applicable.
	M5757/18-001 M5757/18-003	2 units each part number. Qualification inspection table, group I.

(F) Quality conformance inspection: Performance of groups B and C tests are not applicable.

Custodians:
Navy - EC
Air Force - 85

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0779-03)

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